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# A Numerical Investigation of Shear Instability, Mixing and Afterburn Behind Explosive Blast Waves

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## Abstract

A hybrid two-phase numerical methodology is used to study the propagation of explosive blast wave from spherical charges of TNT and their interaction with ambient distribution of aluminum particles. The presence of these particles is found to cause Rayleigh-Taylor instabilities at the contact surface between the detonation products and the shock-compressed air, which results in enhanced mixing and afterburn. The afterburn energy release is not observed to affect the primary blast wave, but affects the pressure decay rate behind the blast wave, and the speed and the strength of the secondary shock. The dispersion and reaction of aluminum particles in the hot region behind the blast wave is investigated for a range of particle sizes and mass loading, and the role played by these particles in the growth of hydrodynamic instabilities is studied. It is shown that for the range of sizes investigated, particle size does not play a significant role in the mixing, but the distribution and initial extent do have appreciable impact. It is found that large particles do not ignite, while small particles rapidly ignite and burn completely. Furthermore, intermediate size particles are observed to ignite only when the mixing is enhanced, however, subsequently quench as they leave the mixing layer. This study has provided some useful insights on the instabilities induced by ambient reactive particles in detonation flow-fields and establishes a simulation capability to study turbulent two-phase processes in an explosive environment.

## I. Introduction

Computational studies can play an important role in understanding the nature of flow dynamics left behind after a blast wave propagates through a medium since experimental in-situ observations are difficult due to the short time-scales of the processes. When a detonation occurs, the outgoing blast wave will heat and put into motion the ambient field as it propagates through it. Simultaneously, a rarefaction wave propagates inwards, forcing an outward acceleration of the detonation product gases. The contact surface between the detonation products and the shock compressed air is swept outwards, and is hydrodynamically unstable to perturbations due to the large density gradients across it. Any disturbance in the ambient flow field or behind the blast wave can be perturbed by these moving shock and expansion waves, causing them to grow further. Shock-shear interactions occur and can result in instabilities that can significantly change the post shock flow fields. Past studies[1] have clearly demonstrated that perturbations can result in hydrodynamic instabilities. These instabilities are primarily Rayleigh-Taylor[2] at early times, and are caused due to the perturbation of the contact surface. If an explosive charge is "hydrodynamically" smooth, instabilities start to grow from the molecular scales. On the other hand, if the surface of an explosive charge is rough, instabilities start to grow from this characteristic scale. In both scenarios, the instabilities grow with time, and result in a

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turbulent mixing layer between the detonation products at the core, and the shock-compressed air, which are separated by the contact surface. The presence of inert/reactive particles, interior or exterior to the charge, can also result in hydrodynamic instabilities, owing to their interaction with the contact surface.

A short while after the detonation event, the inward moving rarefaction wave over-expands the flow, and this gives rise to a secondary shock.[3] This secondary shock is initially weak, and is swept outwards by the detonation product gases. During this time, the secondary shock strengthens, and subsequently implodes inwards. When the secondary shock reflects from the origin, it propagates outwards and interacts with the Rayleigh-Taylor structures, giving rise to further growth of these hydrodynamic instabilities, this time in the form of Richtmyer-Meshkov instabilities.[4] Thus, the flow-field behind explosive blast waves are characterized by hydrodynamic instabilities, that cannot be properly represented by simple one-dimensional models.

The growth of hydrodynamic instabilities has been studied in the past[1,5] by introducing random Gaussian fluctuations at the outer surface of a high explosive charge. Four distinct phases were identified: a blast phase, an implosion phase, a re-shock phase, and an asymptotic mixing phase. The authors also showed the growth of vortical structures by the baroclinic mechanism (misaligned pressure and density gradients), and concluded that most of the afterburn occurs in the asymptotic phase. When reactive (or inert) particles are added to explosives, the detonation product gases heat up the metal particles, and subsequently ignite them. The burning of these particles releases energy, which enhances the blast effects of explosive charges. Afterburn of gram-range charges of PETN surrounded by flake aluminum has been investigated in closed volumes (calorimeters).[6] The authors showed that the mean chamber pressure can be enhanced by such charges, where the high explosive is surrounded by reactive aluminum particles. In another experimental study,[7] confined explosions of kilo-range TNT charges were investigated with aluminum, present either inside the charge, or outside as a surrounding shell. The authors showed that the presence of aluminum contained in a shell outside the charge improved the performance during the early afterburning phase, as compared with charges of the high explosive containing aluminum particles. They also showed that both charges result in the same quasi-static explosion overpressure after a long duration.

Aluminum particles when present along with explosive blasts, pick up heat either from the detonation products, or from the shock compressed air, depending on which side of the contact surface they belong. Once these particles reach their ignition temperature, they start to evaporate, and subsequently react in the gaseous phase. This reaction of gaseous aluminum can be by two different means, i.e., aerobic or anaerobic.[8] Aerobic reaction refers to the reaction of aluminum with the oxygen in the shock compressed air, while anaerobic reaction refers to its reaction with oxidants in the detonation products, viz. $CO_2$ and $H_2O$. It is not possible to determine experimentally, what proportions of aluminum combustion behind blast waves occurs aerobically or anaerobically, and thus, computational simulations can play a useful role in explaining the aluminum combustion process behind explosive blast waves. To this end, this paper is also aimed at the investigation of the burning aspects of aluminum particles behind the explosive blasts.

In this paper we study numerically the detonation of condensed phase explosives with a shell of Al particles outside the charge with a primary focus to investigate the interaction between shock, density interface, particles and the afterburn process. The momentum and heat transfer to the aluminum particles, and their subsequent afterburn as the blast wave propagates outwards is discussed. The impact of these processes on the hydrodynamic instabilities is also addressed. This paper is organized as follows: in Section II, the governing equations and numerical methodology are presented. In Section III, the results from the current study are reported and the involved physics elucidated. Finally, in Section IV, the conclusions drawn upon this research effort are presented.

## II. Governing Equations and Numerical Method

### II.A. Gas Phase

The three-dimensional simulations are conducted using the unsteady, compressible, reacting, multi-species Navier-Stokes equations, and are summarized as:

$$\frac{\partial}{\partial t}\begin{bmatrix}\rho\\ \rho u_i\\ \rho E\\ \rho Y_k\end{bmatrix}+\frac{\partial}{\partial x_j}\begin{bmatrix}\rho u_j\\ \rho u_i u_j+p\delta_{ij}-\tau_{ij}\\ (\rho E+p)\,u_j-u_i\tau_{ji}+q_j\\ \rho Y_k\,(u_j+V_{j,k})\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\\ \dot{\omega}_k\end{bmatrix}+\begin{bmatrix}\dot{\rho}_p\\ \dot{F}_{p,i}\\ \dot{Q}_p+\dot{W}_p\\ \dot{S}_{p,k}\end{bmatrix} \tag{1}$$

where $\rho$ denotes the density, $u_i$ is the i-th component of velocity, $E$ is the specific total energy given by the sum of the internal ($e$) and the kinetic energy, $e + \frac{1}{2}u_i u_i$, $p$ is the pressure, $Y_k$, the mass fraction of the k-th species, and the chemical production of the k-th species is represented by $\dot{\omega}_k$. Denoting the total number of chemical species as $N_s$, the index $k$ in the species equation varies as $k = 1,...,N_s$. The stress tensor is denoted by $\tau_{ij}$, j-direction heat flux by $q_j$, and the j-component diffusion velocity by $V_{j,k}$. The shear stress is obtained as

$$\tau_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) - p\delta_{ij} + \delta_{ij}\lambda\frac{\partial u_j}{\partial x_j}, \tag{2}$$

where $\mu$ represents the viscosity of the gas phase, obtained from the Sutherland's law, $\delta_{ij}$ is the Kronecker delta, and $\lambda$ denotes the coefficient of bulk viscosity, taken as $-\frac{2}{3}\mu$. The heat flux is obtained as

$$q_j = -\kappa\frac{\partial T}{\partial x_j} + \rho\sum_{k=1}^{N_s} h_k Y_k V_{j,k}, \tag{3}$$

where $T$ denotes the temperature, $\kappa$, the thermal conductivity, and $h_k$, the specific enthalpy for the k-th species. The diffusion velocity is computed from Fick's law, i.e., $V_{j,k} = -D_k/Y_k\,(\partial Y_k/\partial x_j)$, where $D_k$ denotes the diffusion coefficient of the k-th species, obtained from unity Schmidt number assumption. The last bracket on the right hand side of Eqn. (1) represents the inter-phase coupling terms, to be discussed shortly.

Following recent studies, we use the Noble-Abel equation of state,[8–10] given by

$$p = \frac{\rho RT}{1 - An} \tag{4}$$

where $R$ denotes the gas constant, $n$, the number of moles per unit volume, and $A$, an empirical constant. The speed of sound ($a$) for a Noble-Abel gas is given by

$$a^2 = \frac{1}{1 - An}\sqrt{\gamma RT} \tag{5}$$

where $\gamma$ denotes the ratio of the specific heats. The Noble-Abel equation of state is chosen for the present study mainly because of its simplicity, and for its transition to the ideal gas equation of state for lower densities. The empirical constant, $A$, is determined from two criterions: (1) ensuring the term $1 - An$ always remains positive; (2) from apriori knowledge of the blast wave overpressure for a detonation case without particles.

The chemical source term, $\dot{\omega}_k$ that arises due to combustion/afterburn has to be determined. Due to the very high temperatures and pressures involved in the problem under study, the conventional finite-rate Arrhenius kinetics based reaction rates are not applicable, as these curvefit expressions are based on very different flow conditions, i.e., relatively lower temperatures and pressures. Moreover, due to the very high pressures involved in the problem under study, the reaction rate can also depend on pressure,[11] which is not well documented. However, the reaction rates for the problem under study are expected to be very fast, and so, in order to simplify the combustion model, the chemical rates are obtained using infinite rate chemistry. This assumption is generally used to model afterburn in explosives, see for instance.[5,8,9] For the present study, we use a six step mechanism:

$$\begin{array}{llllllll}
(1) & C + \frac{1}{2}O_2 \rightarrow CO & & & (4) & Al + O_2 \rightarrow AlO_2 & & \text{if}\, T \geq 3500K \\
(2) & CO + \frac{1}{2}O_2 \rightarrow CO_2 & & & (5) & Al + H_2O \rightarrow AlO + H_2 & & \\
(3) & Al + \frac{1}{2}O_2 \rightarrow AlO & \text{if}\, T < 3500K & & (6) & Al + CO_2 \rightarrow AlO + CO & &
\end{array} \tag{6}$$

The $3^{rd}$ and $4^{th}$ reactions above represent aerobic, while the $5^{th}$ and $6^{th}$ represent anaerobic reaction of aluminum.

The simulation code is an established DNS/LES solver for combustion flows.[12,13] To enable the precise capture of smooth vortical structures, as well as sharp discontinuities (eg. shocks), the current solver combines a central scheme for smooth flow with a shock capturing scheme based on MUSCL reconstruction using the Harten-Lax-vanLeer (HLL) approximate Riemann solver by means of a hybrid HLLC/HLLE approach.[14,15]

### II.B. Detonation Initialization

For the initialization of the detonation profiles within the charge, a one-dimensional simulation is carried out employing the Gas-Interpolated-Solid Stewart-Prasad-Asay (GISPA) method for the detonation process.[16] This method permits time-accurate simulation of detonation from the time of the initial shock through the completion of the explosive burn. The GISPA algorithm is robust, emphasized by its ability to capture the reaction zone as well as the Von Neumann spike. The GISPA method is based upon the reactive Euler equations[17]

$$\frac{\partial \vec{U}}{\partial t} + \frac{\partial \vec{F}}{\partial x} = \vec{S}_G + \vec{S}_{Rx}, \tag{7}$$

where $\vec{U} = (\rho, \rho u, E, \rho\lambda)^T$ is the vector of conserved variables, $\lambda$, the reaction progress state variable, and $\vec{F}$, the flux vector $\vec{F} = (\rho u, \rho u^2 + p, u(\rho E + p), \rho u\lambda)^T$. The source term $\vec{S}_G$ in Eqn. (7) mathematically corrects the one-dimensional equations for non-planar coordinate systems[16]

$$\vec{S}_G = -\frac{j}{x}(\rho u, \rho u^2, u(E+P), \rho u\lambda)^T. \tag{8}$$

where $x$ denotes distance, and $j$ is set to 0 for planar, 1 for cylindrical, and 2 for spherical detonation. The second source term is given by $\vec{S}_{Rx} = (0, 0, 0, r_x)$, and the reaction rate expression $r_x$ can take different forms for different explosives,[18] and thus, has to be chosen appropriately. These equations are solved with the use of appropriate equations of state for both the condensed explosive and the detonation products.[16] For the condensed explosive, the Hayes equation of state is used,[19] while the JWL equation of state is used for the detonation products. The Glaister's[20] version of the Roe scheme is used with MUSCL reconstruction for solving the equations to obtain the initial detonation profile. Figure 1 shows the initial profiles used for the present study.

### II.C. Solid Phase

Lagrangian tracking of a dilute concentration of particles is used in this study, and the drag force acting on a particle is used to compute the particle velocity vector ($u_{p,i}$). The heat transfer to/from a particle from/to the gas is evaluated assuming convection. Under some scenarios, the number of particles to be tracked can be very large, and thus, the concept of parcel is used. Here, a parcel represents a group of particles, each corresponding to the same position, velocity vector, and temperature. Factors like computational cost, available memory, and the desired particle mass loading are used to pre-determine the number of particles per parcel and the total number of parcels. The governing equations for the particle phase are summarized as

$$\frac{dx_{p,i}}{dt} = u_{p,i}, \tag{9}$$

$$\frac{dm_p}{dt} = -\dot{m}_p, \tag{10}$$

$$m_p \frac{du_{p,i}}{dt} = \frac{\pi}{2} {r_p}^2 C_D \rho_g |u_{g,i} - u_{p,i}| \left(u_{g,i} - u_{p,i}\right), \tag{11}$$

$$m_p C_p \frac{dT_p}{dt} = 2\pi r_p \kappa_g Nu \left(T_g - T_p\right) - \dot{m}_p L_v, \tag{12}$$

where $m_p$ is the particle mass, $r_p$ is the particle radius, and $\rho_p$ is the particle material density. Furthermore, $C_D$ represents the drag coefficient and is usually expressed as empirical functions of Reynolds number only for dilute flows.[21] In the heat transfer equation, $C_p$ denotes the specific heat of the solid particle; $\kappa_g$, the thermal conductivity of the gas phase and $L_v$, the latent heat of vaporization. The Nusselt number (Nu) is typically expressed as empirical functions of Reynolds number and Prandtl number in the literature.[22] The system of governing equations for the solid-phase are solved using a $4^{th}$ order Runge-Kutta scheme to obtain the solid particle position vector, velocity vector and temperature. The coupling terms that appear on the right side of the gas phase governing equations (Eqn. 1) are obtained by volume averaging over all the particles/parcels in a finite volume ($Vol$) and are given by

$$\dot{\rho}_p = \frac{1}{Vol}\sum_{1}^{N} n_p \dot{m}_p, \tag{13}$$

$$\dot{F}_{p,i} = \frac{1}{Vol}\sum_{1}^{N} n_p \left[\dot{m}_p u_{p,i} + \frac{\pi}{2} {r_p}^2 C_D \rho_g |u_{p,i} - u_{g,i}| \left(u_{p,i} - u_{g,i}\right)\right], \tag{14}$$

$$\dot{Q}_p = \frac{1}{Vol}\sum_{1}^{N} n_p \left[\dot{m}_p h_v + 2\pi r_p \kappa_g Nu \left(T_p - T_g\right)\right], \tag{15}$$

$$\dot{W}_p = \frac{1}{Vol}\sum_{1}^{N} n_p \left[\dot{m}_p u_{p,i} u_{p,i} + \frac{\pi}{2} {r_p}^2 C_D \rho_g |u_{p,i} - u_{g,i}| \left(u_{p,i} - u_{g,i}\right) u_{p,i}\right], \tag{16}$$

$$\dot{S}_{p,k} = \frac{1}{Vol}\sum_{1}^{N} n_p \dot{m}_p \ (for\, Al\, only), \tag{17}$$

where N is the total number of parcels in a finite volume cell, $n_p$ is the number of particles per parcel, and $h_v$ denotes the enthalpy change associated with the mass transfer.

### II.D. Aluminum Evaporation Model

To model evaporation an empirical quasi-steady law is used:[8,23,24]

$$\frac{dr_p}{dt} = -\frac{r_p}{t_b}\left(1 + 0.276\sqrt{Re}\right) \tag{18}$$

where $Re$ denotes the Reynolds number based on the relative velocity between the gas and the particle, and $t_b$ represents a burning time based on the $d^2$ law, i.e., $t_b = \mathrm{K}\ {d_p}^2$, where $d_p$ represents the particle diamater, and K, an empirical evaporation constant. Thus, the mass transfer rate due to evaporation can be obtained as

$$\frac{dm_p}{dt} = -\dot{m}_p = \frac{d}{dt}\left(\frac{4}{3}\pi \rho_p {r_p}^3\right). \tag{19}$$

It is well known that Al particles exist with an oxide coating, which is generally very thin when compared to the particle diameter. This oxide coating plays an important role in the ignition of the aluminum particles. If the coating does not crack open due to physical and/or thermal stresses, the melting point of aluminum oxide ($\sim 2050$ K) determines the temperature of ignition. On the other hand, if the oxide coating cracks open, the pure (un-oxidized) aluminum is exposed to the outer gases, and thus, the melting point of aluminum ($\sim 950$ K) determines the ignition temperature. Some research studies in the recent past[11,25] have shown that the high gas velocities involved in scenarios similar to the one currently under study, result in the aluminum oxide coating to crack open, and thus, the melting point of aluminum determines the ignition temperature. Some researchers[24] have used 1350 K as the ignition temperature, while others[26] have used a value of 900 K. In the present study, we choose 1000 K as the ignition temperature of aluminum.

## III. Results and Discussion

The aforementioned hydrodynamic instabilities are critical to the analysis of explosive events such as blast waves and supernovae explosions. All features of interest are related to 3D aspects due to the interaction of perturbations and/or particle motion with the density interface. Past studies have confirmed that although the initial propagation speed and shape of the explosion blast wave are not significantly affected by these ambient features, the post shock region has a clear dependence on the same.

In the current study, we use a spherical sector grid approach for the analysis. This approach considers only a part of a sphere, i.e., a spherical sector centered about the equator. The main advantage of using only a part of the sphere is the reduction in computational simulation time. Recently, we used this approach to analyze the propagation and impulsive loading of explosive charges containing dense inert metal particles.[27] This approach has also been used to study turbulent mixing in spherical implosions.[28] However, one of the

problems associated with this approach is the singularity at the origin. For a sector grid, the finite-volume scheme fails near the origin, as the surface area tends to zero. To overcome this singularity at the origin, we assume a small spherical ball at the origin, so that the finite volume scheme is used only beyond a small radial sector around the origin. The size of the spherical ball must be small when compared to the initial charge size, so that the simulated charge contains almost the same amount (e.g., $>$ 99.9 %) of the high explosive as the real charge. A similar approach has been used to study heat transfer in a cone, where a small hemisphere was used around the origin to eliminate the singularity.[29]

The choice of resolution to be used in the radial ($r$), azimuthal ($\theta$) and zenith ($\phi$) directions is crucial to the sector grid approach. Our studies show that a resolution of about 20-25 in the radial direction within the initial charge suffices to accurately capture the blast wave, and an azimuth/zenith resolution of $\Delta\theta = \Delta\phi = 1^o$. For the present study, we use a $2m$ long, and $45^o$ sector, with a resolution of 750X45X45 in $r$, $\theta$ and $\phi$, respectively. Free-slip boundary conditions are used along the sides of the sector and supersonic outflow in the outward plane. The initialization uses the one-dimensional detonation profiles obtained from the GISPA method (Section II.B).

For all the studies, we consider a 11.8 cm dia. spherical TNT charge, and use the sector grid. We study five general cases. Case 1 considers no particles and no fluctuations, i.e., the 1D solution to the problem under study without any role for hydrodynamic instabilities. Case 2 considers without any particles but with Gaussian random fluctuations added at the outer surface of the charge. A similar approach has been used by other researchers,[1,5] albeit they added the fluctuations outside the charge, while we add them in the region $0.9r_o \leq r \leq r_o$, where $r_o$ denotes the charge radius (= 5.9 cm in this case). In both these scenarios, the perturbations are added to trigger the growth of instabilities. When the perturbations are inside the initial charge, the leading blast wave does not interact with them, and thus, the initial structures start to grow in the Rayleigh-Taylor mode. On the other hand, when the initial perturbations are outside the charge, the leading blast wave has to propagate through these perturbations, and thus, the early growth is Richtmyer-Meshkov in nature. In the current study, the perturbations can be assumed to arise from imperfections in the explosive material.

Case 3 considers three different cases of large aluminum particles (radius 250 $\mu$m, 100 $\mu$m, and 50 $\mu$m), all of a fixed mass loading, $\eta = 1$, randomly arranged immediately outside the charge until a radius of 0.25 m. Here, mass loading is defined as the ratio of particle to gas mass in a control volume at initial time. In terms of volume fraction, it is still in the dilute range and is 0.044% for $\eta = 1$. No random fluctuations are added, as these particles serve as the initiator/trigger of hydrodynamic instabilities. Case 4 considers three different cases of intermediate to small aluminum particles (radius 50 $\mu$m, 10 $\mu$m, and 5 $\mu$m), also for a fixed mass loading, $\eta = 1$, randomly arranged immediately outside the charge until a radius of 0.0868 m. Finally, Case 5 considers a fixed aluminum particles size (50 $\mu$m radius), but of varying mass loading ratios, $\eta = 0$, 0.25, and 1, randomly arranged immediately outside the charge until a radius of 0.25 m. With these studies we attempt to explore the effect of mass loading, particle size, initial perturbation and afterburn on the overall flow features. Table 1 summarizes the different cases considered in this study.

**Table 1. Cases considered in the present study**

| Case | Particle radius, $\mu$m | mass loading ratio ($\eta$) | radial extent of particles, m |
|---|---|---|---|
| 1 | no particles | 0 | no fluctuations |
| 2 | no particles | 0 | Gaussian fluctuations |
| 3 | 250, 100, 50 | 1 | 0.059-0.25 |
| 4 | 50, 10, 5 | 1 | 0.059-0.0868 |
| 5 | 50 | 0, 0.25, 1 | 0.059-0.25 |

## III.A. Gas phase

For the rest of the study, we use Case 2 as the baseline.

### *III.A.1. Effect of perturbation*

Figure 2 shows the mixing layer (iso-surface of $N_2$ mass fraction) shape at four different times. During the initial blast wave phase, the structures grow in time, yet preserving their initial perturbation shape as evident

from the profile at 0.4 msec (Fig. 2 (a)). A mixing layer is created where the detonation products and the shocked air co-exist. Since the detonation products and the shocked air mix for the first time, afterburn occurs, resulting in energy release. During the implosion phase, the secondary shock, as it implodes inwards, drags along with it the lower end of the mixing layer, as evident from the profile at 1 msec (Fig. 2 (b)). More afterburn occurs during this phase, as the detonation products and the shocked air mix. During the re-shock phase, the secondary shock passes through the mixing layer from the inside, a classic Richtmyer-Meshkov scenario, resulting in more vorticity creation due to baroclinic effects ($\dot{\omega} = -\nabla(1/\rho) \times \nabla p$), which in turn leads to further mixing enhancement in the layer as evident from the profile at 2 msec (Fig. 2 (c)). This baroclinic effect will be revisited and discussed in detail later for Case 3. Subsequently, in the asymptotic phase, contiguous structures begin to interact and merge, thereby giving rise to a more distorted and wrinkled apperance to the mixing layer as observed from the profile at 8.5 msec (Fig. 2 (d)). This merging between structures results in loss of memory of the initial perturbation shape.

To understand the effect of afterburn, the pressure traces at the radial location 0.9 m from the charge center are plotted for the studies involving one-dimensional mixing-free (Case 1) and three-dimensional with mixing (Case 2) cases for the same TNT charge in Fig. 3. The increased mixing and afterburn associated with the three-dimensional case is not observed to affect the primary shock, as the afterburn energy release occurs over a time frame of a few hundred milli-seconds, which is not fast enough to couple with the primary shock. However, the secondary shock is observed to be slightly faster and stronger for the three-dimensional case, due to the higher speed of sound in the mixing region caused by the increased afterburn energy release, which in turn results in lesser attenuation rate of the secondary shock as it traverses through the mixing region. The decay rate of the pressure profiles behind the primary shock is also different between 1D and 3D studies. For instance, at around 1 msec, for the case with instabilities and enhanced mixing/afterburn, the pressure decay is less than the corresponding 1D. Thus, the mixing and afterburn energy release are associated with three important features: (1) acceleration of the secondary shock; (2) stronger secondary shock, and (3) slower decay rate of the pressure behind the primary shock.

### *III.A.2. Effect of particle size*

The pressure-time profiles for Cases 3 and 4 are shown in Fig. 4 at the 0.9 m radial location, along with the particle-free baseline profile. As observed, the primary shock strength and arrival time is nearly the same implying that the Rayleigh-Taylor instability induced mixing and the afterburn energy release are too slow to couple with the primary shock. This is in agreement with the observations for Case 2 (Fig. 3). There are, however, some differences between the particle laden cases and the baseline case. To see this more carefully, we show an expanded view of the pressure trace in Fig. 5. In Fig. 5 (a) & (b), the pressure-time trace at the 0.9 m radial location is shown for Cases 3 and 4, respectively, along with the particle-free case. The decay behind the primary shock is less for Cases 3 and 4 compared with the baseline case (see for instance around 1.25 msec). Furthermore, the secondary shock arrival time is earlier for Cases 3 and 4 than for the particle-free case, emphasizing the role of particles in the mixing process. The presence of particles perturbs the contact surface to a considerable extent, and allows for the growth of Rayleigh-Taylor instabilities, which in turn results in enhanced mixing between the detonation products and the shock-compressed air. Enhanced mixing directly contributes to enhanced afterburn energy release, which in turn decreases the pressure decay rate behind the primary shock wave for the particle laden cases. Moreover, the afterburn energy release increases the speed of sound in the mixing layer, thereby resulting in a faster secondary shock.

Analysis shows that the Rayleigh-Taylor instability caused by the interaction of the contact surface with the particles starts to grow from a scale comparable to the particle size. At later times, the nonlinear growth of the instability loses memory of the initial conditions regardless of the initial particle size, and all cases for different particle sizes, but with the same initial radial extent give rise to approximately the same pressure field behind the primary blast wave. Thus, although the instabilities may begin from a scale comparable to the particle size, the flow physics (entrainment, turbulence, etc.) dictates subsequent mixing and afterburn. These observations are consistently seen in many other simulations (not included here for brevity) and suggests an overall similarity of the flow field in the later stage of detonation. From a practical point of view, this is an important observation since it suggests that the late stage observation of the flow field may not directly correlate to what occurred at the initial stage of detonation propagation in gaseous or particle laden flow-field. However, further study is still needed to fully quantify these observations.

In order to better understand the growth of instability to the initial conditions, the region of particle distribution is varied systematically in Cases 3 and 4. For the 50 $\mu$m radius particles, the pressure-time

traces are compared for different initial extent of particle distribution, 0.25 m and 0.0868 m in Fig. 5 (c) (Cases 3 and 4), along with the baseline particle-free case. In this case, the pressure behind the primary shock decays less and the secondary shock is faster when the initial particle distribution extends farther. Since there are more particles in number (for the same loading ratio), the initial triggering of instabilities from the contact surface lasts for a longer time (and distance) for Case 3 when compared to Case 4 for the same particle size (50 $\mu$m). Due to this longer interaction period, mixing and hence, the afterburn is increased resulting in less decay of the pressure and a faster secondary shock for the 50 $\mu$m size in Case 3.

The iso-surface of the oxygen mass fraction corresponding to 50% of the ambient oxygen mass fraction is shown in Fig. 6 at four dfferent times for the 10 $\mu$m particle size considered in Case 4. The structures grow at early times, and subsequently, after the passage of the secondary shock, the structures start to merge due to baroclinic torque effects and the ensuing vorticity generation. This results in a loss of memory of the shape of these structures at subsequent times.

### *III.A.3. Baroclinic torque*

The baroclinic torque and vortex stretching phenomena are critical to the problem under study. The vorticity equation for a compressible flow is summarized as

$$\frac{D\vec{\omega}}{Dt} = \left(\vec{\omega}.\vec{\nabla}\right)\vec{V} - \vec{\omega}\left(\vec{\nabla}.\vec{V}\right) + \frac{1}{\rho^2}\vec{\nabla}\rho \times \vec{\nabla}p + \vec{\nabla} \times \left(\frac{\vec{\nabla}.\tau}{\rho}\right), \tag{20}$$

where $\vec{\omega}$ is the vorticity vector. The first term on the right side denotes the vortex stretching due to velocity gradients; the second term denotes the vortex stretching due to flow compressibility, and the third term is the baroclinic torque. The last term denotes the diffusion of vorticity due to viscous effects. The baroclinic term creates vorticity whenever the density and pressure gradients are mis-aligned. The two vortex stretching terms contain $\vec{\omega}$, indicating that these terms come into play only if a non-zero vorticity already exists. A vortex subjected to a parallel velocity gradient is bound to stretch, due to which its cross-section decreases. In order to conserve angular momentum, the vorticity magnitude has to increase. Thus, this term is critical in the mixing layer behind blast waves, where vortical structures are subjected to large velocity gradients. Furthermore, during the passage of the secondary shock through the mixing layer, the latter is observed to compress (also reported elsewhere[1]), and thus, the vortex stretching due to flow compressibility is also critical in the mixing layer.

For the problem under study, during the re-shock phase, the secondary shock passes through the mixing layer, a classical Richtmyer-Meshkov scenario.[4] During this phase, the secondary shock interacts with the structures present in the mixing layer, giving rise to baroclinic-torque due to the mis-alignment of pressure and density. Hence, the vorticity increases during the re-shock phase due to baroclinic effects, and the vortical structures grow due to the vortex stretching terms, i.e., due to velocity gradients and flow compressibility. Figure 7 shows these three terms: (a) $ln\left(1 + \left|\left(\vec{\omega}.\vec{\nabla}\right)\vec{V}\right|\right)$; (b) $ln\left(1 + \left|\vec{\omega}\left(\vec{\nabla}.\vec{V}\right)\right|\right)$ , and (c) $ln\left(1 + \left|\frac{1}{\rho^2}\vec{\nabla}\rho \times \vec{\nabla}p\right|\right)$ at 1.4 msec for the 50 $\mu$m particle size in Case 3. It is observed that the baroclinic term increases as the secondary shock interacts with the structures in the mixing layer, but subsequently decays. This baroclinic term creates vorticity, which in turn results in vortex stretching as the vortical structures interact with velocity gradients and flow compressibility. Due to this, both the vortex stretching terms are observed to sustain even after the passage of the secondary shock, as observed by the "red regions" behind the secondary shock for both the stretching terms (Figs. 7 (a) & (b)). Furthermore, all these three terms are observed to be of a similar magnitude, thereby indicating that they are all equally important during the re-shock phase. Due to the vorticity creation during the re-shock phase, the secondary shock is observed to distort in shape, however re-attaining a spherical shape after leaving the mixing layer, not shown here for brevity. The contiguous structures observed to interact in Figs. 2 & 6 at the later times is a result of this vorticity creation, which further enhances mixing between the inner detonation products and the outer air.

### *III.A.4. Effect of mass loading*

Here, we analyze the effect of the mass loading ratio ($\eta$) to the problem under study. We present the pressure-time traces for the 50 $\mu$m particle size initially extending until 0.25 m radius, but with loading ratios $\eta = 0$, 0.25 and 1 in Fig. 8. As observed, the pressure decay rate is less, and the secondary shock faster, when the loading ratio is higher, i.e., when more particles are available to perturb the contact surface.

### III.B. Solid phase

*III.B.1. Particle cloud trajectory*

The trajectories of the leading and trailing edges of the particle clouds are shown for the three different particle sizes considered in Case 3, along with the primary shock trajectory in Fig. 9. Since the momentum transfer time scales are proportional to ${d_p}^2$, the particle cloud trajectories are very different for the three different sizes considered. In all these cases, the blast wave gains the initial lead over the particle cloud, but then attenuates due to its spherical nature. The particles, owing to their inertia, decelerate lesser than the gas, and thus catch up with the blast wave and subsequently overtake it. The radial distance and time duration at which the particle cloud overtakes the blast wave is different for the different sizes considered. The 250 $\mu$m particle cloud overtakes the blast wave at around 1.1 m radial location (Fig. 9 (a)). Also, the particle cloud width (gap between the leading and trailing edges of the particle cloud) is observed to widen rapidly with time for the 250 $\mu$m case, due to the slower relaxation times. For the 100 $\mu$m case, the leading edge of the particle cloud overtakes the blast wave earlier, i.e., at around 0.65 m radial location, due to the relatively shorter momentum transfer time scale (Fig. 9 (b)). Also for the same reason, the cloud width grows with time at a slower rate. For the 50 $\mu$m case, due to the even shorter momentum transfer time scales, the leading edge of the particle cloud overtakes the blast wave almost immediately (Fig. 9 (c)). Subsequently, due to aerodynamic drag, the leading particles are slowed down and are thus, observed to maintain a constant lead of about 4 cm ahead of the blast wave, unlike the other larger sizes considered, which get farther ahead of the blast wave after overtaking it. Furthermore, for the 50 $\mu$m case, the cloud width is almost constant with time due to the faster momentum transfer time scales. The overtaking of the blast wave by the particles is better illustrated in Fig. 10 for the 250 $\mu$m case at times 0.3 and 0.84 msec, the particles are shown as black dots against the background contour of $ln\ (\rho)$. At the earlier time, the particles lag the blast wave, but have overtaken it at the later time.

For the 10 $\mu$m size in Case 4, the particles quickly attain momentum from the gas and are set into motion, attaining velocities in excess of 2000 m/s by 0.2 msec. By 0.53 msec, all the particles have slowed down to about 1000 m/s, and to about 600 m/s by 0.98 msec, the rapid aerodynamic deceleration being due to the smaller momentum transfer time scales.

*III.B.2. Particle ignition/quenching*

The particles corresponding to the 250 and 100 $\mu$m cases did not reach the ignition temperature due to their large size and the associated large heat transfer time scales. For the 50 $\mu$m size in Case 3, most aluminum particles reached the ignition temperature within about 0.3 msec, thereby starting to evaporate. However, as the gas temperature drops in the region between the detonation products and the vicinity of the leading blast wave, the particles that ignited are quenched within about 0.8 msec as the surrounding gas temperature drops significantly. During this short time, the ignited particles undergo partial vaporization, and the gaseous aluminum is observed to combust both aerobically and anaerobically to form oxides of aluminum. To illustrate this ignition and subsequent quenching, the particle radius and temperatures are shown at 0.3 and 0.96 msec for the 50 $\mu$m size in Case 3 in Fig. 11, with $ln\ (\rho)$ as the background contour. As observed, the maximum particle temperature at 0.96 msec is 903 K, which is lower than the ignition temperature. The smallest particle radius is 46.7 $\mu$m at 0.3 msec, and is 24.2 $\mu$m at 0.96 msec, showing partial vaporization.

The 50 $\mu$m size considered in Case 4 did not ignite, unlike in Case 3 for the same size, thus emphasizing that for intermediate particle sizes, the afterburn energy release is critical to ignite the particles. For a wider initial extent of the particle distribution (Case 3), the mixing and afterburn are enhanced, and thus the chances of particle ignition are also enhanced for the same particle size (50 $\mu$m) and mass loading ratio ($\eta$ = 1). For the 5 and 10 $\mu$m sizes considered in Case 4, it is observed that the particles start to evaporate very quickly ($<$ 0.04 msec), unlike the 50 $\mu$m size considered in Case 3, not shown here for brevity. Since the 5 & 10 $\mu$m sizes have very small heat transfer time scales, the outgoing blast wave heats up the particles beyond the ignition temperature, i.e., the particles do not depend on the afterburn of the detonation products to supply the additional heat to ignite, as observed for the 50 $\mu$m particles in Case 3. Furthermore, all the 5 $\mu$m sizes are completely evaporated by about 0.3 msec.

To illustrate the evaporation rates of solid aluminum, the time varying solid aluminum mass remaining for four cases (refer Table 2) which involve vaporization are shown in Fig. 12 normalized with the total initial mass of the aluminum particles. As evident, the 50 $\mu$m particles corresponding to $\eta$=1 (case 1) ignite

at around 0.2 msec, and evaporate almost linearly in time until they get quenched around 0.8 msec. By decreasing the loading ratio (case 4), the 50 $\mu$m ignite later, viz. at 0.4 msec, and quench earlier, viz. at 0.7 msec. Thus, higher loading ratios give rise to enhanced mixing, and enhanced afterburn of the detonation products, which is observed to significantly affect the ignition and quenching characteristics of the aluminum particles. The 10 $\mu$m particles (case 2) burn more slowly than their 5 $\mu$m (case 3) counterparts, and the latter evaporate completely by 0.3 msec. This has some practical implications: since the 5 $\mu$m aluminum particles burn very efficiently, they can be used in conjunction with the larger particles, thereby supplying energy to assist the ignition of the latter. However, these studies are beyond the scope of this paper and will not be addressed here.

**Table 2. Cases shown in Fig. 12**

| Case | $r_p$, $\mu$m | loading ratio ($\eta$) | radial extent of particles, m |
|---|---|---|---|
| 1 | 50 | 1 | 0.059-0.25 |
| 2 | 10 | 1 | 0.059-0.0868 |
| 3 | 5 | 1 | 0.059-0.0868 |
| 4 | 50 | 0.25 | 0.059-0.25 |

From the above simulations, the following observations can be summarized for the gas phase: (i) ambient particles perturb the contact surface giving rise to hydrodynamic instabilities; (ii) mixing and afterburn are significantly affected by the presence of hydrodynamic instabilities; the pressure field behind the blast wave is (iii) independent of particle size for the same mass loading ratio and initial radial extent, (iv) enhanced by the width of the initial particle distribution for the same particle size and mass loading ratio, and (v) enhanced by a higher loading ratio for the same particle size and initial radial extent. For the solid phase, the relevant conclusions are: (i) clouds of larger particles take a longer time to catch-up to the primary blast wave, (ii) cloud width for larger particles grow with time, but asymptotes to a constant width for smaller particle sizes, (iii) very large particles ($\geq$ 100 $\mu$m) do not ignite and very small particles ($\leq$ 10 $\mu$m) are immediately ignited by the blast wave, (iii) intermediate particle size (50 $\mu$m) is not ignited by the blast wave alone, and depend on the afterburn energy to ignite, and (iv) intermediate particle size quench at subsequent times.

## IV. Conclusions

A robust hybrid numerical methodology is used to study the propagation of explosive blast waves from a spherical TNT charge into reactive aluminum particles. The presence of these particles is found to cause Rayleigh-Taylor instabilities at the contact surface between the detonation products and the shock-compressed air, which results in enhanced mixing and afterburn. The problem under study is characterized by four distinctive phases, consistent with earlier particle-free studies in which the instabilities were triggered by other means. During the outward passage of the secondary shock through the mixing layer, Richtmyer-Meshkov instabilities are observed, and vorticity is significantly enhanced by the baroclinic torque effect. This creation of vorticity is found to be responsible for contiguous structures to interact, thereby resulting in a loss of memory of the initial shape of the instabilities.

The afterburn energy release also significantly affects the flow-field behind the blast wave. The post-detonation afterburn of the detonation products is observed to be enhanced by the mixing process, and significantly affects the flow-field behind the blast wave. The pressure decay rate behind the primary blast wave is lowered, and the secondary shock is observed to be faster and stronger due to the presence of hydrodynamic instabilities.

The motion, heating and burning characteristics of the aluminum particles is also investigated and explained. While large particle did not ignite, small particles rapidly ignite and burn completely. Intermediate size particles are found to ignite only when the mixing is enhanced, but are found to subsequently quench as they leave the mixing layer. This study has provided some useful insights on the flow-fields behind explosive blast waves and the instabilities induced by ambient reactive particles.

## V. Acknowledgments

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## References

[1]Kuhl, A. L., "Spherical Mixing Layers in Explosions," *Dynamics of Exothermicity, Gordon and Breach Publishers*, 1996, pp. 291–320.

[2]Taylor, G. I., "The Instability of Liquid Surfaces when Accelerated in a Direction Perpendicular to their Planes," *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, Vol. 201, 1950, pp. 192–196.

[3]Brode, H. L., "Blast Wave From a Spherical Charge," *The Physics of Fluids*, Vol. 2, 1959, pp. 217–229.

[4]Richtmyer, R. D., "Taylor Instability in a Shock Acceleration of Compressible Fluids," *Communications on Pure and Applied Mathematics*, Vol. 13, 1960, pp. 297-319.

[5]Kuhl, A. L., Ferguson, R. E., and Oppenheim, A. K., "Gasdynamic Model of Turbulent Exothermic Fields in Explosions," *Progress in Astronautics and Aeronautics*, Vol. 173, 1997, pp. 251–261.

[6]Kuhl, A. L., Neuwald, P., and Reichenbach, H., "Effectiveness of Combustion of Shock-Dispersed Fuels in Calorimeters of Various Volumes," *Combustion, Explosion and Shock Waves*, Vol. 42, No. 6, 2006, pp. 731–734.

[7]Zhang, F., Anderson, J., and Yoshinaka, A., "Post-Detonation Energy Release from TNT-Aluminum Explosives," *AIP Shock Compression of Condensed Matter, Waikoloa, Hawaii*, 2007.

[8]Kim, C. K., Moon, J. G., Hwang, J. S., Lai, M. C., and Im, K. S., "Afterburning of TNT Explosive Products in Air With Aluminum Particles," *AIAA Paper 2008-1029*, 2008.

[9]Schwer D. A., and Kailasanath, K., "Numerical Simulation of the Mitigation of Unconfined Explosions Using Water-Mist," *Proceedings of the Combustion Institute*, Vol. 31, 2007, pp. 2361–2369.

[10]Johnston I. A., "The Noble-Abel Equation of State: Thermodynamic Derivations for Ballistics Modelling," *Defence Science and Technology Organisation*, DSTO-TN-0670, 2005.

[11]Kim, K., Wilson, W., Peiris, S., Needham, C., Watry, C., Ortley, D. J., and Zhang, F., "Effects of Particle Damage During Detonation of Thermobarics on Subsequent Reactions," *International Colloquium on the Dynamics of Explosions and Reactive Systems, Poitiers, France*, 2007.

[12]Menon, S., and Patel, N., "Subgrid Modeling for Simulation of Spray Combustion in Large-Scale Combustors," *AIAA Journal*, Vol. 44, No. 4, 2006, pp. 709–723.

[13]Eggenspieler, G., and Menon, S., "Large-Eddy Simulation of Pollutant Emission in a DOE-HAT Combustor," *Journal of Propulsion and Power*, Vol. 20, No. 6, 2004, pp. 1076–1085.

[14]Genin, F., and Menon, S., "Simulation of Interaction of Shocks With Turbulent Shear Layers Using Large-Eddy Simulations," *Computers and Fluids*, Under preparation, 2008.

[15]Balakrishnan, K., Genin, F., Menon, S., and Nance, D. V., "Numerical Study of Propagation Characteristics from Detonations in Homogeneous Explosives," *To be submitted to Shock Waves*, 2008.

[16]Xu, S., Aslam, T. and Stewart, D. S., "High Resolution Numerical Simulation of Ideal and Non-ideal Compressible Reacting Flows with Embedded Internal Boundaries," *Combustion Theory and Modeling*, Vol. 1, 1997, pp. 113–142.

[17]Xu, S. and Stewart, D. S., "Deflagration-to-detonation Transition in Porous Energetic Materials: a Comparative Model Study," *Journal of Engineering Mathematics*, Vol. 31, 1997, pp. 143–172.

[18]Wescott, B. L., Stewart, D. S. and Davis, W. C., "Equation of State and Reaction Rate for Condensed Phase Explosives," *Journal of Applied Physics*, Vol. 98, 2005, 053514.

[19]Hayes, D. B., "A $P^n t$ Detonation Criterion from Thermal Explosion Theory," *Sixth Intrenational Symposium on Detonation, Pasadena, CA*, 1976.

[20]Glaister, P., "An Approximate Linearized Riemann Solver for the Euler Equations for Real Gases," *Journal of Computational Physics*, Vol. 74, 1998, pp. 382–408.

[21]Clift, R., Grace, J. R., and Weber, M. E., *Bubbles, Drops and Particles*, Academic, New York, 1978.

[22]Drake, R. M., "Discussion on the Paper Entitled 'Forced Convection Heat Transfer From an Isothermal Sphere to Water' by G. C. Violet and G. Leppert," *ASME Journal of Heat Transfer*, Vol. 83, 1961, pp. 170–172.

[23]Khasainov, A., and Veyssiere, B., "Steady, Plane, Double-front Detonations in Gaseous Detonable Mixtures Containing a Suspension of Aluminum Particles," *Dynamics of Explosions: Progress in Astronautics and Aeronautics*, Vol. 114, 1988, pp. 284–299.

[24]Benkiewicz, K., and Hayashi, A. K., "Parametric Studies of Aluminum Combustion Model for Simulations of Detonation Waves," *AIAA Journal*, Vol. 44, No. 3, 2006, pp. 608–619.

[25]Gilev, S. D., and Anisichkin, V. F., "Interaction of Aluminum with Detonation Products," *Combustion, Explosion and Shock Waves*, Vol. 42, 2006, pp. 107–115.

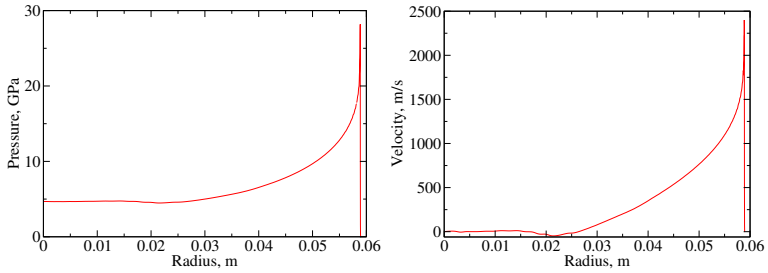
[26]Fedorov, A. V., Fomin, V. M., and Khmel, T. A., "Theoretical and Numerical Study of Detonation Processes in Gas Suspensions with Aluminum Particles," *Combustion, Explosion and Shock Waves*, Vol. 42, No. 6, 2006, pp. 735–745.

[27]Balakrishnan, K., and Menon, S., "Simulation of Blast Wave Propagation and Particle Motion from Detonation Containing Dense Inert Particles," *AIAA Paper 2008-4689*, 2008.

[28]Youngs, D. L., and Williams, R., "Turbulent Mixing in Spherical Implosions," *International Journal for Numerical Methods in Fluids*, Vol. 56, 2008, pp. 1597–1603.

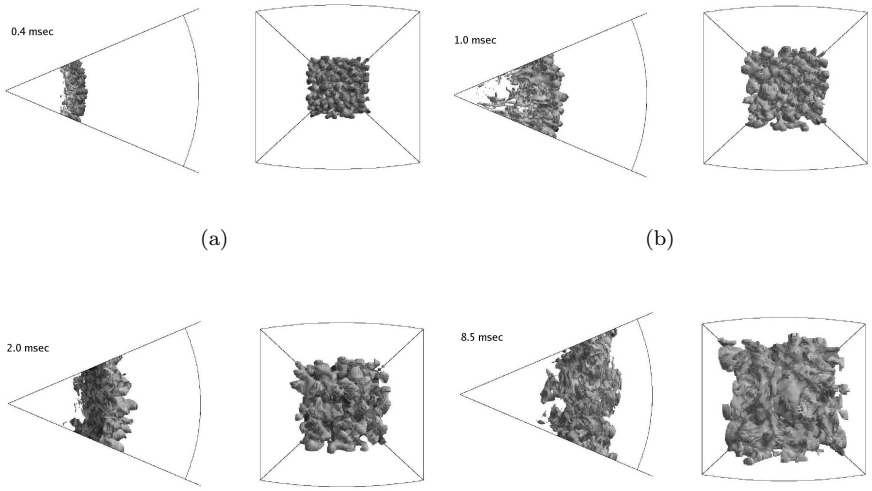
[29]Pickard, W. F., "Base to Apex Thermal Conductance of a Cone Embedded in a Non-ideal Insulator," *International Journal of Heat and Mass Transfer*, Vol. 47, 2004, pp. 5271–5283.

# VI. Figures




(a) (b)

**Figure 1. Initial detonation profiles for the TNT charge (a) pressure and (b) velocity.**




(a) (b)

(c) (d)

**Figure 2. Mixing layer growth at times (a) 0.4 msec; (b) 1 msec; (c) 2 msec and (d) 8.5 msec.**

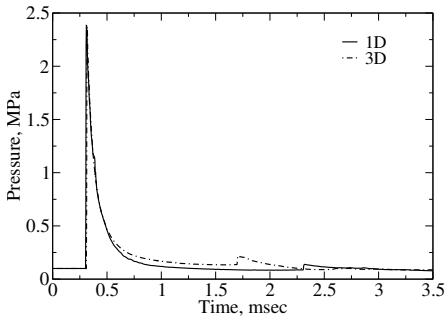



**Figure 3. Pressure traces at the 0.9 m radial location for the TNT charge.**

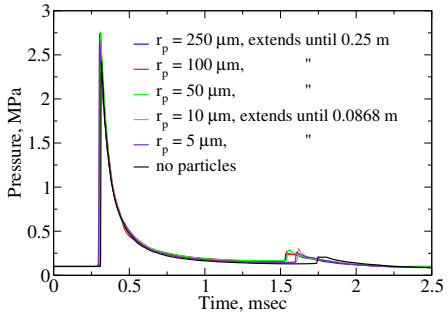
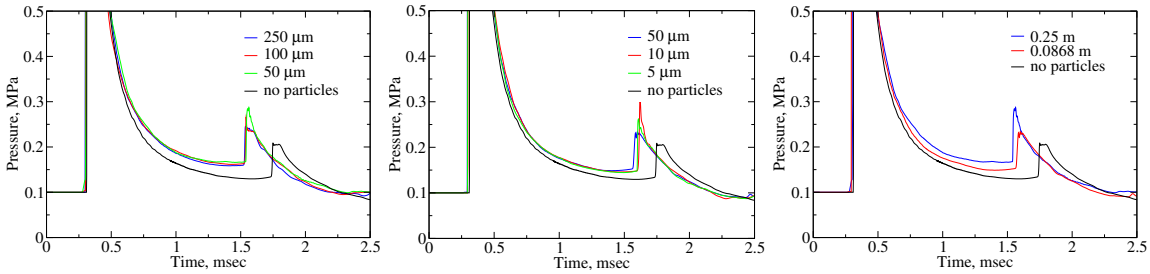



**Figure 4. Pressure profiles at the 0.9 m radial location for Cases 3 & 4.**




(a) (b) (c)

**Figure 5. Effect of mixing and afterburn on the pressure profiles.**

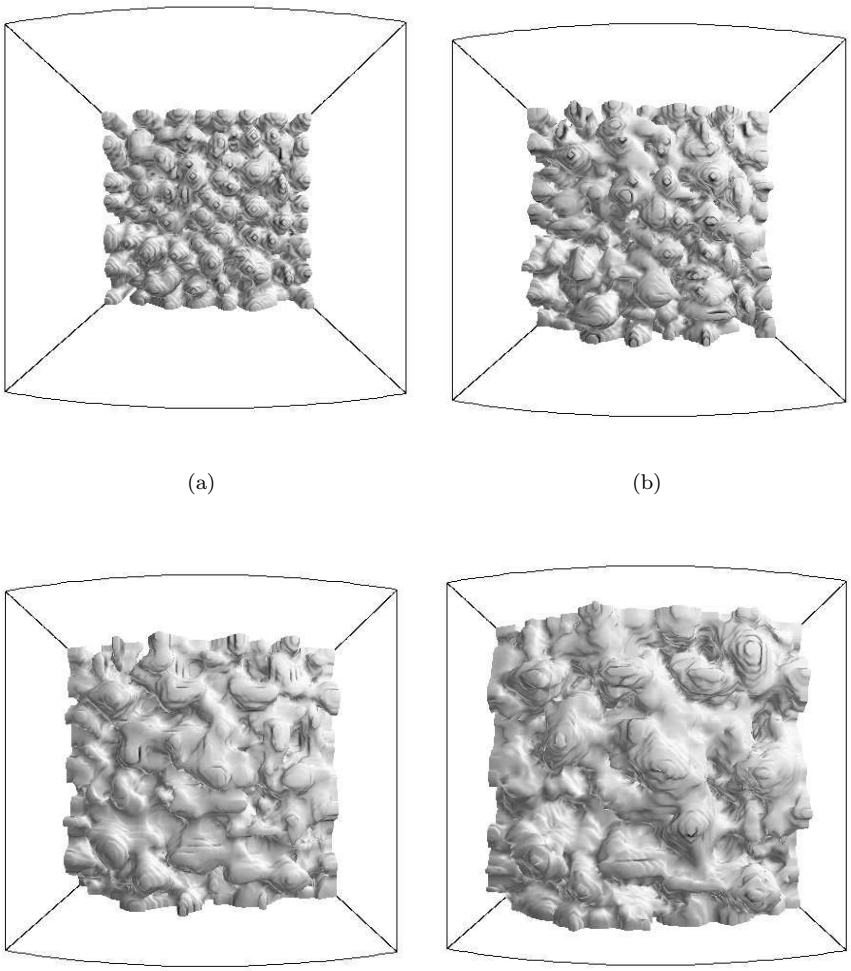



**Figure 6. Iso-surface of oxygen mass fraction to qualitatively illustrate the mixing process at times (a) 0.53 msec, (b) 1.14 msec, (c) 2.09 msec and (d) 3.68 msec.**

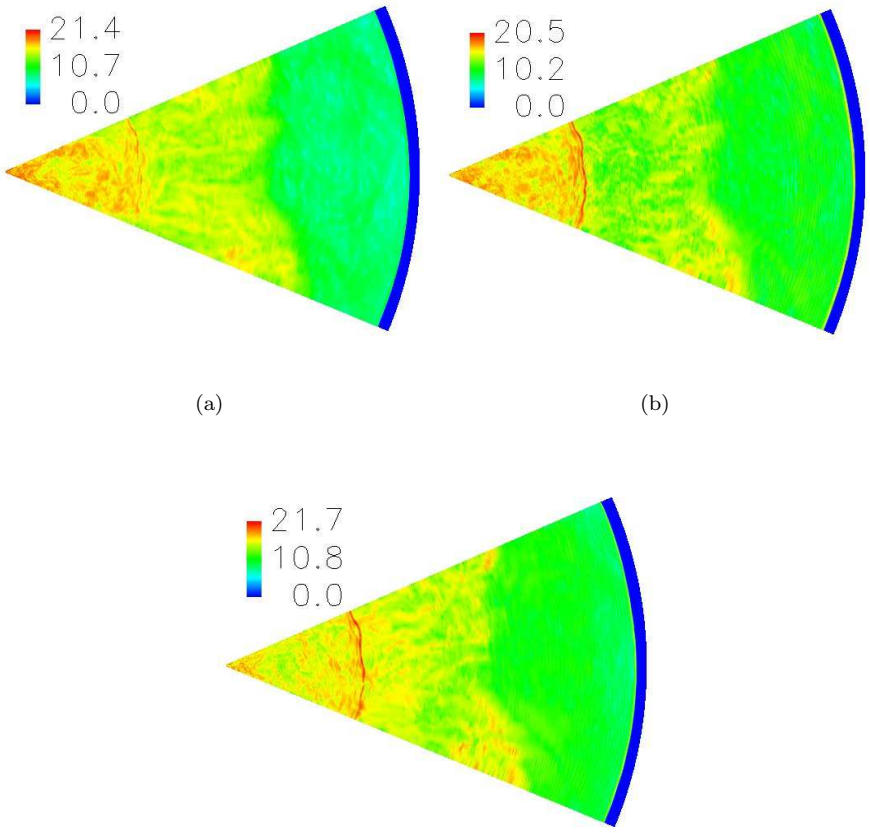



**Figure 7. Re-shock phase at 1.4 msec: (a) vortex stretching due to velocity gradients; (b) vortex stretching due to flow compressibility, and (c) baroclinic term.**

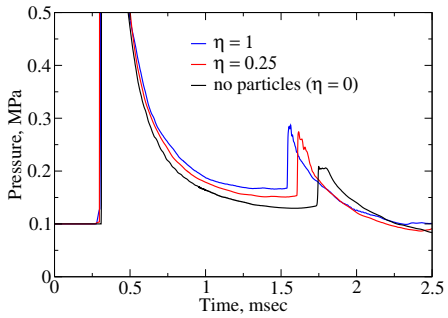
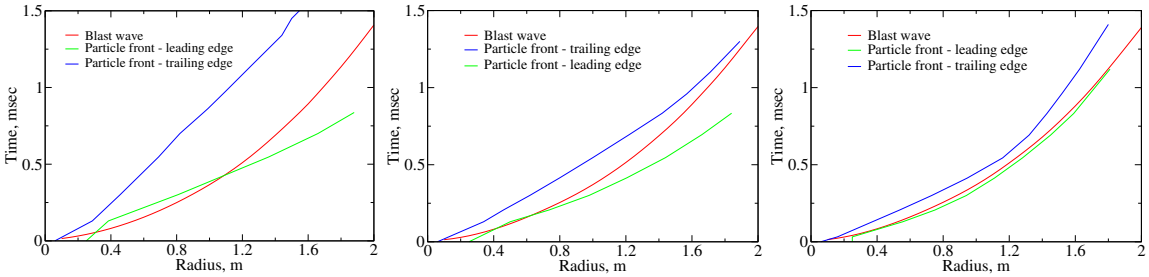
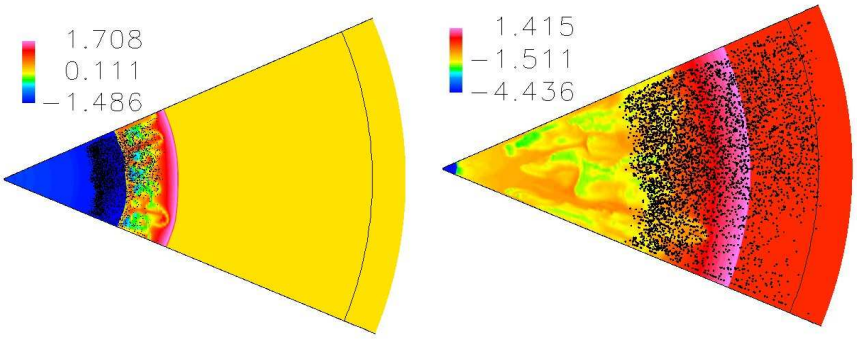



**Figure 8. Effect of loading ratio on the pressure-time trace.**



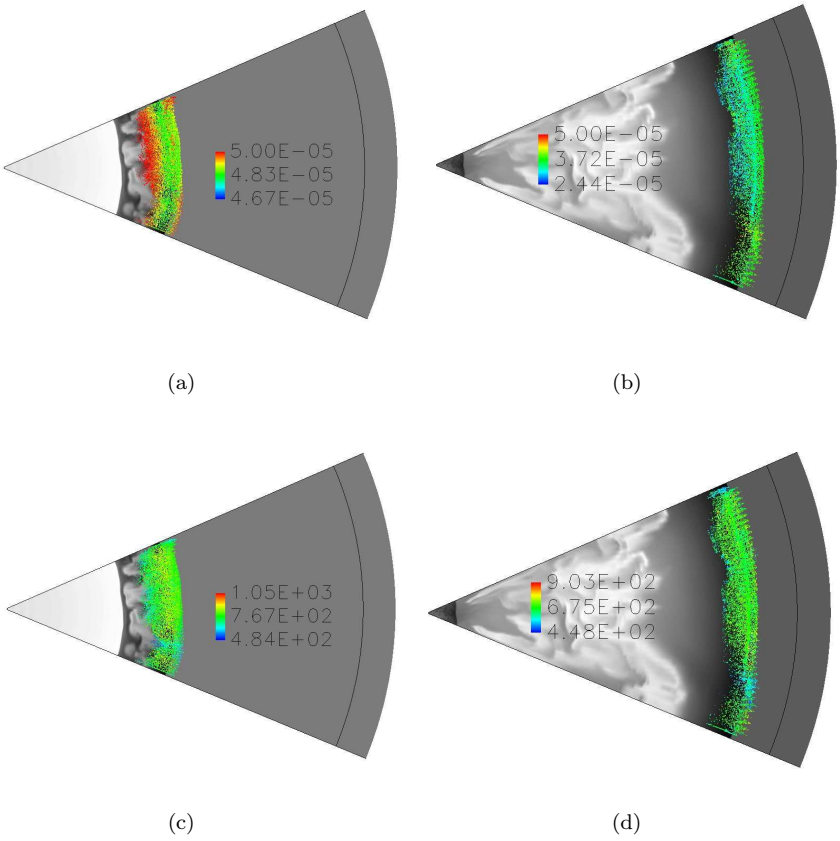

(a) (b) (c)

**Figure 9. Particle cloud trajectories for Case 3: (a) 250 $\mu$m, (b) 100 $\mu$m and (c) 50 $\mu$m.**




(a) (b)

**Figure 10. Particle cloud for the 250 $\mu$m case in Case 3: (a) 0.3 msec, (b) 0.84 msec.**




(a) (b)

(c) (d)

**Figure 11. Particle radius (a & b) and temperature (c & d) for the 50 $\mu$m case at times 0.3 msec (a & c) and 0.96 msec (b & d).**

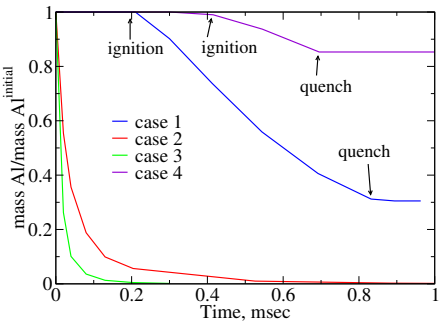



**Figure 12. Solid aluminum mass remaining with time.**